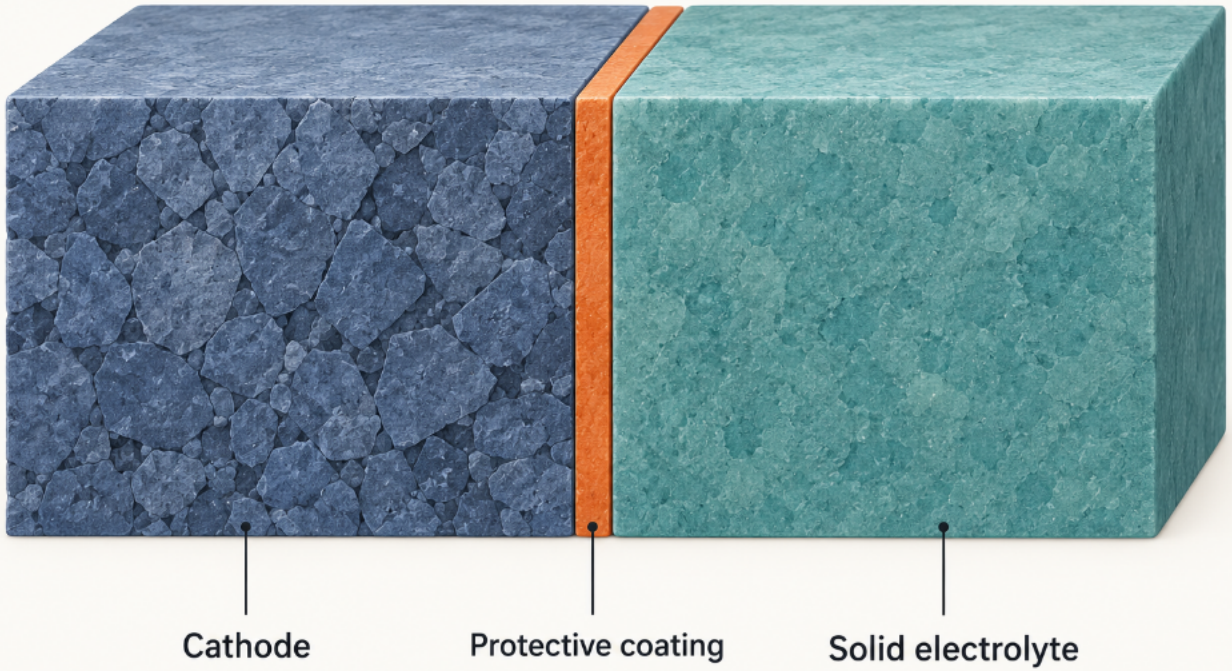
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interphAIse: choosing protective battery layers

An AI language model chooses what to test. Scientific tools check its choices.



Conceptual illustration, not to scale

A coating must

Limit unwanted reactions and stay attached.

Block electrons.
Let lithium ions cross.

Goal: test whether feedback improves coating selection.

Davide Tisi and Adem R. N. Aouichaoui

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The LLM and its scientific tools

Choose a candidate, calculate its properties, read the result, choose again.

Part of the workflow	Tool and role
Scientific choices	Haiku 5.5 / DeepSeek V3.2 Select a known material and explain the choice
Chemical screening	Materials Project + pymatgen / SciPy Supply reference data and precompute reaction scores
Physical follow-up	ASE + PET-MAD Model the atoms and calculate attachment

141 known crystal structures representing 100 compositions.

Three electrolytes. Ten choices per search. 810 completed searches.

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Five strategies test the value of feedback

Thirty searches per strategy. Different runs may find the same material.

Strategy	How it chooses
Uses results	One choice at a time, with the true result after each test
No results	One at a time, remembering choices but seeing no scores
Mixed-up results	One at a time, with scores assigned to the wrong materials
One list	All ten choices together, before seeing any result
Random	Software samples ten materials without repeats

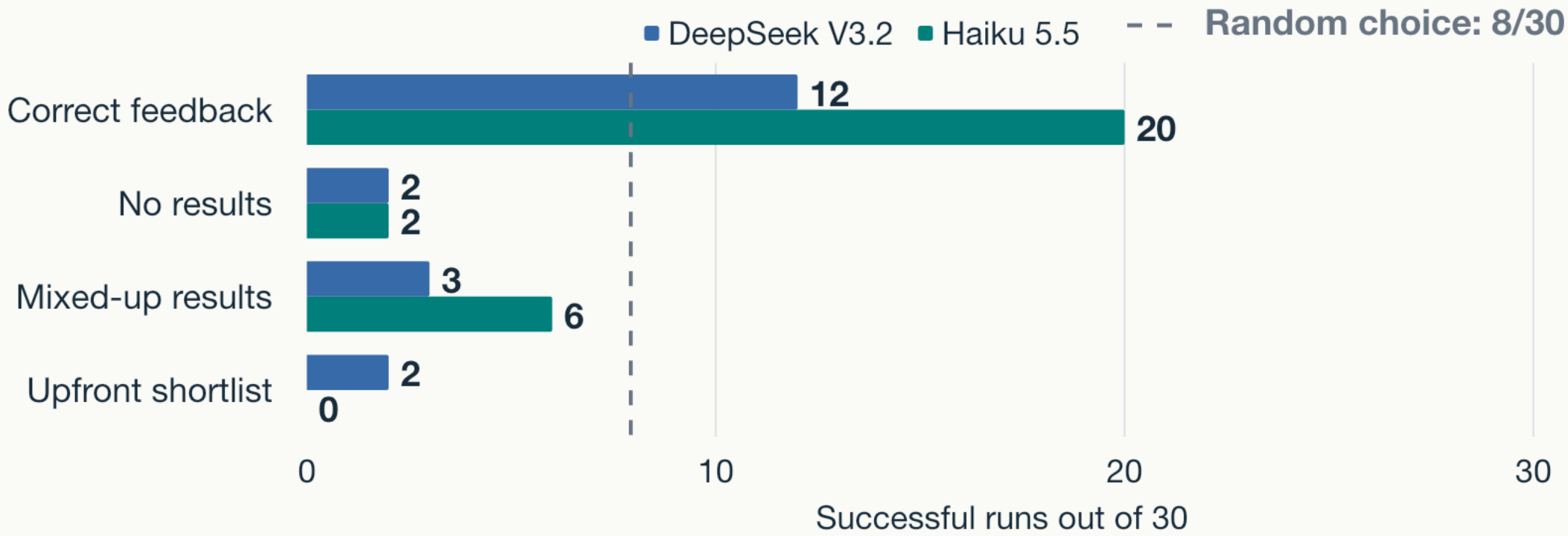
Pass = penalty ≤ 0.050 eV/non-Li atom AND band gap ≥ 2 eV.

Search success = at least one pass within ten choices.

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Sulfide: correct feedback helps Haiku

Success = at least one accepted candidate within ten choices. Each strategy has 30 runs.



Correct feedback: Haiku 20 and DeepSeek 12, compared with random 8.

Haiku beats both controls after correction. DeepSeek remains inconclusive.
Cross-model contrasts are descriptive. Reasoning compute differs.

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Sulfide follow-up: why these coatings?

Haiku selects and forecasts. PET-MAD tests both contacts.

Coating	Why it was chosen	Weaker-contact adhesion (J/m²)
$LiNbF_6$	Existing acceptable reference	1.633
$LiTaF_6$	Acceptable, close Nb analogue	1.577
$LiCaAlF_6$	Only acceptable option left	0.744

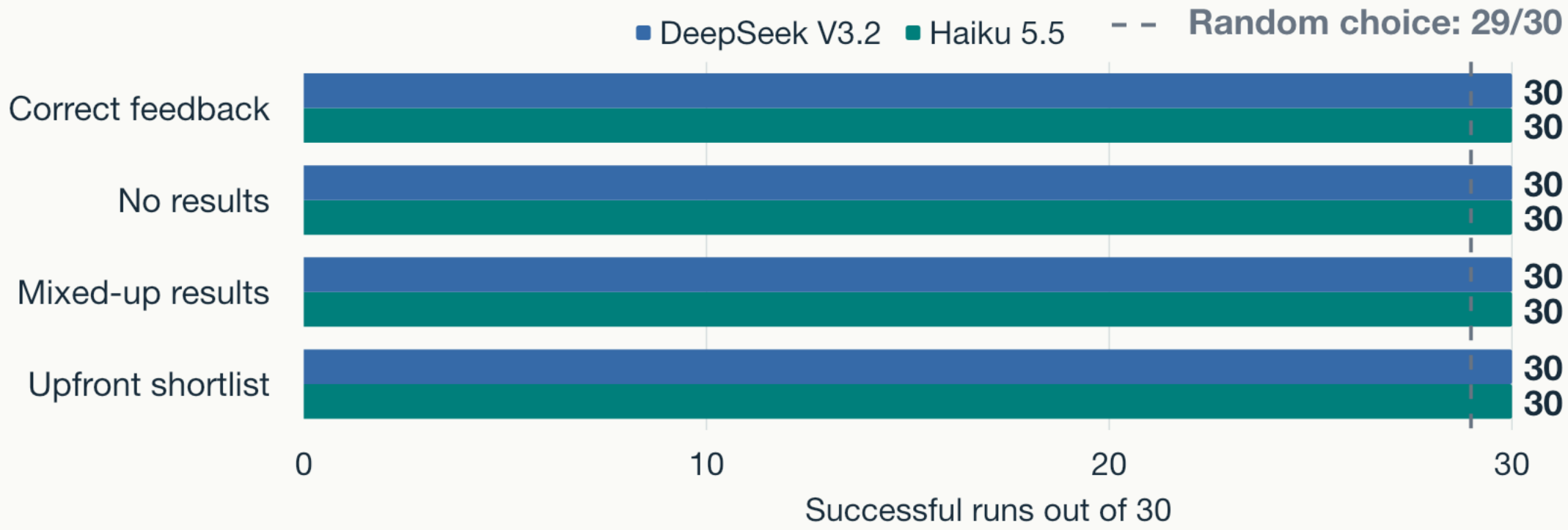
Ta forecasts held. Ca–Al electrolyte forecast: 1.4–1.9 J/m².
PET-MAD returned 0.744 J/m². The forecast failed.

Provisional adhesion from formally polar interfaces. Lithium transport is untested.

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Chloride: almost every strategy succeeds

Success = at least one accepted candidate within ten choices. Each strategy has 30 runs.



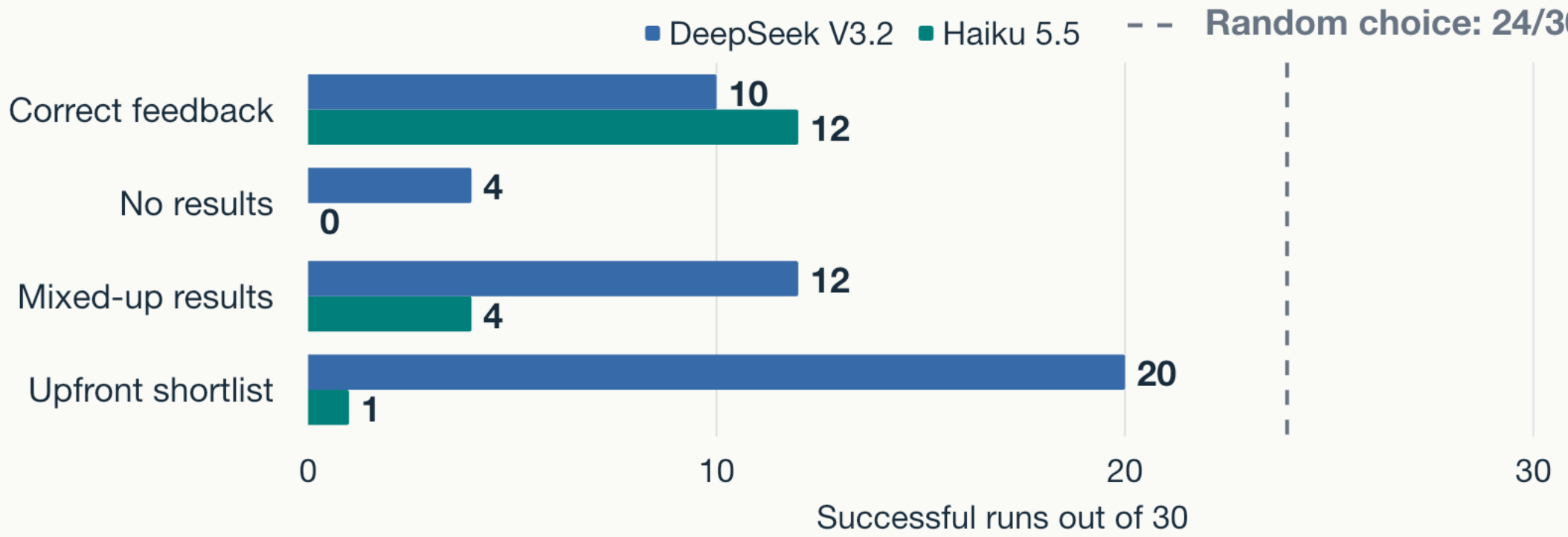
All LLM strategies: 30/30. Random choice: 29/30.

55 of 141 structures pass. This easy screen leaves little room for a feedback advantage.

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LLZO: the feedback loop falls short

Success = at least one accepted candidate within ten choices. Each strategy has 30 runs.



Every LLM strategy trails random choice on the observed success count.

DeepSeek: 20 upfront, 10 with feedback. This also changes the selection policy.
The acceptance proxy admits questionable oxyhalogens. The frozen counts stay unchanged.

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What the LLM contributes

Useful chemical knowledge and useful adaptation to feedback are distinct capabilities.

Next: test lithium transport through the shortlisted coatings.

Then validate with higher-accuracy calculations and experiments.
No human-expert comparison or validated coating yet.

Code and evidence: github.com/DavideTisi/interphAIse

Structure viewer: davidetisi.github.io/interphAIse